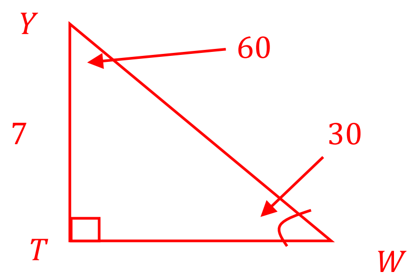
Geometry
Unit 7
Lesson 6 Homework

Name_Answer Key_________
Date____________________
Period___________________

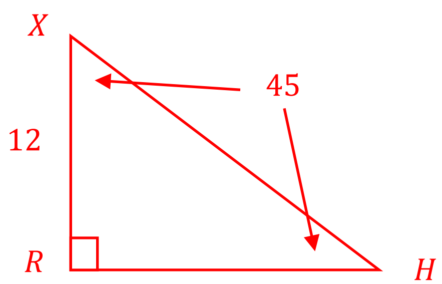
$\triangle YWT$ is a right triangle where $m\angle YTW = 90°$, $m\angle YWT = 30°$, and $YT = 7$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
1.	$\overline{WT}$	$7\sqrt{3}$
2.	$\overline{YW}$	14



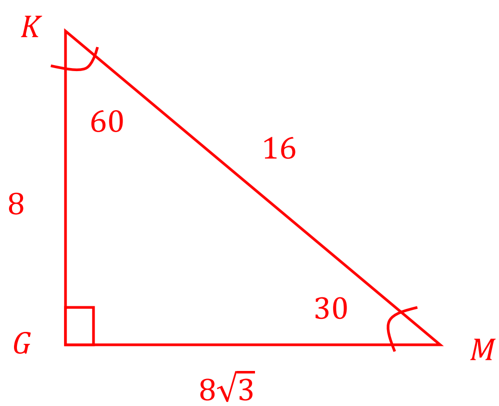
$\triangle RXH$ is a right triangle where $m\angle XRH = 90°$, $m\angle RXH = 45°$, and $XR = 12$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
3.	$\overline{RH}$	12
4.	$\overline{XH}$	$12\sqrt{2}$



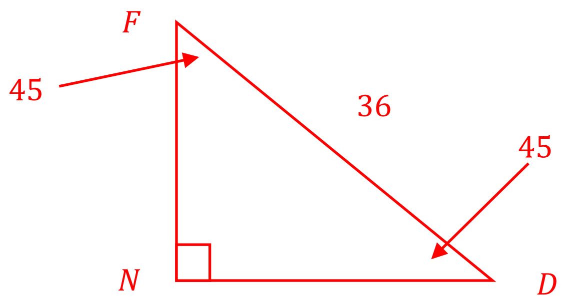
$\triangle KGM$ is a right triangle where $m\angle MGK = 90°$, $m\angle GKM = 60°$, and $KM = 16$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

	Side	Length
5.	$\overline{KG}$	8
6.	$\overline{GM}$	$8\sqrt{3}$



$\triangle FND$ is a right triangle where $m\angle FND = 90°$, $m\angle NDF = 45°$, and $DF = 36$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
7.	$\overline{ND}$	$18\sqrt{2}$
8.	$\overline{NF}$	$18\sqrt{2}$



$\triangle YBM$ is a right triangle where $m\angle YMB = 90°$, $m\angle MYB = 30°$, and $YB = 24$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
9.	$\overline{MB}$	12
10.	$\overline{YM}$	$12\sqrt{3}$